

Practice Quiz 04D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 04

Comparing the power set of a union with a union of power sets.

1. Let $A = \{0\}$ and $B = \{1\}$.

- (a) List all the elements of $\mathcal{P}(A) \cup \mathcal{P}(B)$ and of $\mathcal{P}(A \cup B)$.

Solution We have $\mathcal{P}(A) = \{\emptyset, \{0\}\}$ and $\mathcal{P}(B) = \{\emptyset, \{1\}\}$. Therefore

$$\mathcal{P}(A) \cup \mathcal{P}(B) = \{\emptyset, \{0\}, \{1\}\}.$$

Since $A \cup B = \{0, 1\}$,

$$\mathcal{P}(A \cup B) = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}. \quad \blacksquare$$

- (b) Exhibit a set that belongs to exactly one of the two families in part (a). Explain why it belongs to one and not to the other, and use this to decide whether $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$ always holds.

Solution The set $\{0, 1\}$ belongs to $\mathcal{P}(A \cup B)$ because it is a subset of $A \cup B$. It belongs to neither $\mathcal{P}(A)$ nor $\mathcal{P}(B)$: it is not a subset of A because $1 \notin A$, and it is not a subset of B because $0 \notin B$. Thus it does not belong to their union. These sets give a counterexample, so the proposed equality does not always hold. $\blacksquare$